

Stratospheric Ballooning Association

Low-Cost Software Defined Radio Solutions for High Altitude Applications

Abstract

LoRa (Long Range) and LoRaWAN (Long Range Wide Area Network) are complementary technologies designed to enable low-power, long-range wireless communication for Internet of Things (IoT) applications. LoRa operates at the physical layer using Chirp Spread Spectrum (CSS) modulation, which encodes data using chirp signals—linear frequency sweeps that offer resilience to noise and Doppler effects. The spreading factor (SF) and bandwidth (BW) are critical parameters in LoRa modulation: increasing the spreading factor improves range and noise immunity but decreases data rate, whereas increasing bandwidth allows for higher data rates at the cost of receiver sensitivity. LoRaWAN operates at the MAC (Media Access Control) and network layers, providing standardized communication protocols and secure data exchange. Understanding how data rate impacts bandwidth and system performance is essential for optimizing throughput, battery life, and range in real-world deployments. Software-defined radios (SDRs) are commonly used to evaluate and monitor these signals. Devices like the RTLSDR offer an ultra-low-cost solution suitable for fundamental spectral analysis but are limited in frequency range and transmit capability. In contrast, the Analog Devices PlutoSDR and the HackRF One by Great Scott Gadgets provide more extensive frameworks: the PlutoSDR supports full-duplex operation and is tightly integrated with ADI's libiio framework, offering high flexibility for embedded development. The HackRF, while half-duplex, offers broad frequency coverage and robust community support. Price and architectural flexibility further differentiate these SDR platforms, influencing their suitability for LoRa/LoRaWAN experimentation and deployment.

Software-Defined | Radio | LoRa | LoRaWAN | RTL-SDR | Pluto | SDR | HackRF |

1. Introduction

High-altitude ballooning (HAB) presents a unique set of challenges for researchers, hobbyists, and commercial operators alike. These challenges can generally be categorized into three primary areas: environmental factors (such as atmospheric conditions and weather variability), structural and material constraints (including payload weight, cost, and mechanical integrity), and control and navigation difficulties (such as reliable positioning and communication systems). Cost and communication reliability are often the most critical considerations for sustained and scalable HAB operations.

This paper addresses these concerns by examining LoRa (Long Range) and LoRaWAN (Long Range Wide Area Network) communication protocols. LoRa offers low-power, long-range wireless communication through Chirp Spread Spectrum (CSS) modulation, while LoRaWAN provides a standardized network protocol for managing data exchange and security at higher layers. These technologies are increasingly adopted for HAB applications due to their favorable range balance, power efficiency, and affordability. The paper will parallel evaluate the question: can software-defined radio (SDR) platforms be used to analyze, develop, and deploy LoRa and LoRaWAN systems? Specifically, it will compare three off-the-shelf SDR devices from a practical and economic perspective: the RTL-SDR, a popular entry-level receiver; the Analog Devices ADALM-PLUTO SDR (PlutoSDR), known for its full-duplex operation and developer-friendly framework; and the HackRF One by Great Scott Gadgets, valued for its broad frequency coverage and active community support. Each device will be assessed regarding system flexibility, technical capabilities, and cost-effectiveness, particularly within HAB communication requirements and other practical research applications.

2. What is LoRa (LoRaWAN)?

2.1 LoRa (Long-Range) Wireless Modulation

LoRa, or Long-Range radio modulation, is a proprietary modulation technique developed by SEMTECH Corp. This corporation develops many different types of devices, such as IoT systems, semiconductors, and, as noted here, radio devices and techniques. This technique is based on CSS to enable long-distance wireless communication at a power-conservative method of encoding data packets at lower bit rates via chirp pulses. The versatile nature of the LoRa modulation scheme at both sub-gigahertz frequencies, such as the 433 MHz, the 868 MHz, and the 915 MHz frequency bands, respectively, allows a highly capable modulation of a range of data packet sizes, as seen in Figure 1. Even the 2.4 GHz frequency band allows for higher data rates than the sub-GHz bands, though a range loss should be accounted for at these frequencies. These frequencies are great for amateur balloonists and research involving students, as they fall under the ISM (Industrial, Scientific, and Medical) spectrum, which does not require a license to operate under, allowing for accessibility. [1 - 2]

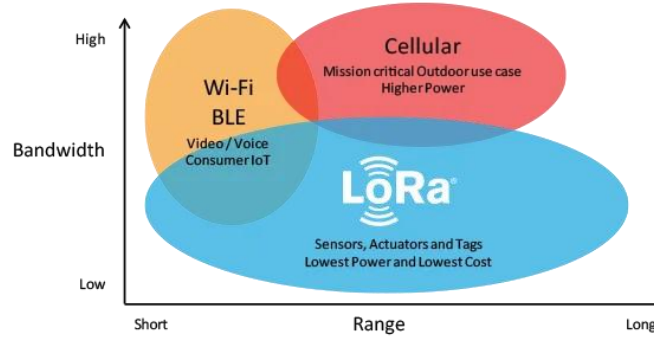


Figure 1: A Bandwidth Vs. Range plot, noting overlap with some HF transmission methods and range against LoRa, and some applications.

<https://www.thethingsnetwork.org/docs/lorawan/what-is-lorawan/>

2.2 LoRaWAN

LoRaWAN, or LoRa Wide-Area Network, is a protocol built for LoRa modulation that works as a MAC layer on top of the LoRa P.H.Y. (Physical Layer) to define how a particular device using the LoRa technique can, for example, format messages or when they can transmit over wireless communication lines. This protocol was developed and maintained by the LoRa Alliance, with the first specification released in January 2015 and the most recent version, version 1.0.4, published in October 2020. LoRaWAN uses a star topology, meaning it uses end devices to send short uplink at rates set by the spreading factor and the bandwidth, which are relayed by either a single or multiple gateways to a central network server. The MAC manages frequencies, data rates, and power settings to optimize power usage and range. [1 - 3]

2.3 Why LoRa?

LoRa modulation has a robust nature; the wideband chirp signals resist in-band interference and Doppler shifts, and it uses strong forward error correction (FEC) to recover signals when the signal-to-noise ratio is below the noise floor, or 0 dB (decibels). The bitrate for LoRa per Hz of bandwidth is incredibly small, i.e., the bitrate for a spreading factor of 7 (see section 3.2) at a bandwidth of 125 kHz is only about 0.044 – 0.055 bps/Hz, which can be calculated with Eq. (1):

$$R_b = \frac{1}{T_s} \log_2 M = \frac{SF}{T_s} = B \frac{SF}{2^{SF}} \quad (1)$$

Where $BT_s = M = 2^{SF}$, with SF as an integer, and $T_s = 2^{SF} \cdot T$ is the symbol interval, $T = \frac{1}{B}$ is the sample rate, and $B = \frac{M}{T_s} = R_c$ is bandwidth. Using this equation, one can see how LoRa can accommodate large link budgets of over 150 dB and provide a capability of around 3 km in densely populated areas and over 10 km in rural areas using LoRaWAN gateways without further help from external equipment such as power (PAs) and low-noise amplifiers (LNAs), which could boost those distance, but may cause the introduction of noise when using PAs on the transmitting end. Due to the ability of the LoRa system, however, this noise should cause little issue when decoding the signal at the receiver while still providing a signal much stronger than Wi-Fi or cellular connections. [4 - 5]

3. C.H.I.R.P., Spread Factor, & Bandwidth

3.1 C.H.I.R.P. & C.S.S.

Chirp, or a linear frequency sweep, and the chirp spread spectrum have been mentioned previously, but what do they mean? Chirp is an acronym for "Compressed High-Intensity Radar Pulse," a signal in which frequency changes over time, also known as symbols. This is what allows it to be used in a spread spectrum. Initially developed for radar applications, CSS uses the complete bandwidth to transmit signals. As in Figure 2, chirps, or frequency changes moving upwards (up-chirp) or downwards (down-chirp) within the frequency band, which is done in a method called Frequency Shift Chirp Modulation (FSCM), a term used by Dr. Lorenzo Vangelista, an associate professor of Telecommunications within the Department of Information Engineering of the University of Padova in Padua, Italy. [5 - 6]

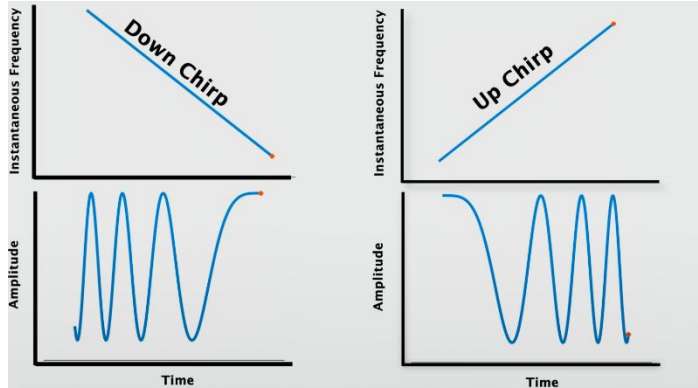


Figure 2: Visual Electric describes how the instantaneous frequency and amplitude of a CSS chirp relate in the video: "[How LoRa Modulation really works - long range communication using chirps](#)" (00:01:04-00:01:28)

3.2 Spreading Factor, Data Rate, & Bandwidth

To go into details about FSCM, the Spreading Factor (SF) must first be defined. The SF is an integer that determines the chirp rate, which controls the data transmission rate. The faster the chirp, the higher data transmission rates can reach, which relates to a lower SF. The higher the SF, the longer the chirp is maintained, lowering the bitrate but increasing processing gain and sensitivity.

From the perspective of LoRa, SF is usually taken as an integer vector from SF7 to SF12, where SF7 is the highest data rate with the shortest chirp and, therefore, lowest range, and SF12 is twice the chirp length, which halves the bitrate but increases link budget for better sensitivity. To calculate how many bits per symbol (M) are found in Eq. (1). If the symbol set $s(nT_s) = 0, 1, 2, \dots, 2^{SF} - 1$, given from Eq. (2):

$$s(nT_s) = \sum_{h=0}^{SF-1} w(nT_s)_h \cdot 2^h. \quad (2)$$

Where $w(nT_s)$ is a vector of SF binary digits. The waveform of time T_s for a given $s(nT_s)$ can be calculated with Eq. (3) and Eq. (4):

$$c(nT_s + kT) = \frac{1}{\sqrt{2^{SF}}} e^{-j2\pi[(s(nT_s)+k) \bmod 2^{SF}]kT \frac{B}{2^{SF}}} \quad (3)$$

$$= \frac{1}{\sqrt{2^{SF}}} e^{-j2\pi[(s(nT_s)+k) \bmod 2^{SF}] \frac{k}{2^{SF}}} \quad (4)$$

for $k = 0 \dots 2^{SF} - 1$.

Eqs. (3) and (4) show the modulated signal as a chirp waveform that increases linearly with k, also known as the time index. Each waveform measurement is taken after the base waveform initial measurements with a frequency equal to 0 differ by a frequency shift son, which is where the term FSCM is used. [5 - 7]

Time-on-air is another factor to consider what observing SF as it grows exponentially with each step on the SF. With each +1 step, the bitrate is halved for a fixed bandwidth or coding rate (CR). This also influences power consumption and receiver sensitivity. Inversely, doubling the bandwidth will double the bitrate a fixed SF and CR as well. An example of this is shown on *Table 1* below:

Table 1: A table presenting bitrates calculated with SF7 and a CR=1 for three different bandwidths (in kHz).
<https://www.thethingsnetwork.org/docs/lorawan/spreading-factors/#:~:text=The%20spreading%20factor%20controls%20the,data%20transmission%20rate%20is%20halved>

Spreading Factor	Bandwidth	Bit rate (kbits/s)
7	125	5.5
7	250	10.9
7	500	21.9

Another table, *Table 2*, shows SF can alter receiver sensitivity. Note the higher the SF is, the lower the sensitivity value for the receiver reception becomes. For H.A.B. applications, battery life also becomes a major issue as power consumption is highly dependent on the SF as well due to longer active time frames for radio transceivers, causing higher power draw and, in turn, shorter battery life.

Table 2: Receiver sensitivity is observed at each of the 6 Spreading Factor points and compared to the fixed bandwidth of 125 kHz, with signal strength in dBm.
<https://www.thethingsnetwork.org/docs/lorawan/spreading-factors/#:~:text=The%20spreading%20factor%20controls%20the,data%20transmission%20rate%20is%20halved>

Spreading factor	Receiver sensitivity for bandwidth fixed at 125 kHz
SF7	-123 dBm
SF8	-126 dBm
SF9	-129 dBm
SF10	-132 dBm
SF11	-134.5 dBm
SF12	-137 dBm

Sensitivity can also be roughly calculated using the noise power density -174 dBm/Hz, the noise power per Hertz of bandwidth, then scaling the noise power with the bandwidth. The sum will provide the received signal strength indicator (RSSI) value, as in *Eq. (5)*.

$$-174 \text{ dBm/Hz} + 10 \log(B) \cong \text{Sensitivity at Receiver.} \quad (5)$$

Because of this, narrower channels, such as B = 125 kHz, with a given power, will have much better reception at longer distances. As seen in Table 2, an SF12 at this bandwidth gives a sensitivity of around -137 dBm, which enables a long-range transmission, though these settings

present a low spectral efficiency. This efficiency can be found by taking the value R_c and divided by the value of E_q (1), as below in E_q (6):

$$\eta = \frac{R_c}{R_b} = \frac{B}{R_b} = B \frac{SF}{2^{SF}} \quad (6)$$

The reciprocal of the value for E_q 6, $\frac{1}{\eta}$, will yield a value in bps/Hz. At SF12, the spectral efficiency is roughly 0.003 bps/Hz, contrasted with higher bandwidths with a lower SF, such as 500 kHz at SF7, providing greater than 20 kbps at the cost of sensitivity. The previously noted resistance to interference and significant processing gain makes this a worthy exchange. Using LoRaWAN's Adaptive Data Rate feature to assign nodes the highest data rate dynamically, each node can carefully plan duty cycles under regulations. [5 - 7]

4. Performance Impacts

4.1 Energy

Constant current draw is often utilized on LoRa-capable devices, regardless of the Spreading Factor. The previously mentioned time-on-air (see section 3.2) becomes the primary concern in power conservation. Also, as previously noted, higher SF implies longer chirp durations, equating to 5-10 times the packet transmission time at SF12 at 125 kHz compared to channels with SF7 at the same bandwidth, causing the device to consume considerably higher mA. In LoRaWAN networks, where the networks use the lowest SF for the highest data rate, the gateway or server often controls each device's SF and power to balance energy usage and transmission range. [8-9]

4.2 Link Budget

The link budget, a calculation used to determine the power at the receiver to account for losses over the transmission channel, directly relates to the capable range of the system and is set by transmit power and receiver sensitivity. The average range of LoRa-capable modems is between +14 to +22 dBm. Using the previous high-SF and narrow bandwidth settings as an example, sensitivity remains around -137 dBm, which is highly dependent on three factors: bandwidth, spreading factor, and noise figure. This would give a link budget value range from 155-170 dB. Changing the bandwidth raises the noise floor at a 1:1 ratio for each +3 dB incrementation, which causes the sensitivity to decrease if the SF is not updated. The forward error correction also negatively affects the signal-to-noise ratio threshold, which is typically between -6 to -20 dB SNR required for LoRa. [2], [8]

4.3 Efficiency

Due to the method in which chirps are formed via LoRa, the spectral efficiency of the modem tends to be on the lower side. However, the processing gains are less susceptible to interference from narrowband transmissions when compared to other systems like FSK modulation. This comes from the asynchronous nature and a high BT value, making it resistant to in- and out-of-band interference. Semtech notes that the symbol period can be significantly more extended than typical short-burst, fast hopping FHSS (Frequency-Hopping Spread Spectrum) systems, and is most resistant to pulsed AM interference. In-band, strong tone interference mechanisms tend to result in upwards of 10 dB loss for average FSK systems, whereas Semtech reports losses of only 1-2 dB in sensitivity. [2], [8]

5. Software-Defined Radio for LoRa/LoRaWAN

LoRa/LoRaWAN research and experimentation often use low-cost SDR hardware for flexible transmission and reception. This section compares three popular SDRs: the RTL-SDR dongle, Analog Devices' ADALM-PLUTO (PlutoSDR), and the HackRF One. These devices vary in frequency range, TX capability, duplex mode, and ecosystem support.

First, what is software-defined radio? The previous explanation of LoRa and LoRa WAN already described how general radio and this particular modulation type and networking compare to other modulation types. When acquiring the equipment, LoRa modems are strictly hardware-based, meaning this hardware sets all specifications. Frequency ranges, filters, amplification, and modulation are all derived from a singular setting or even a narrow range of settings without modifying the hardware. This takes specific electronics skills and the purchase of even more hardware. This lack of flexibility is not nearly as considerable of an issue when using software defined radio equipment, as it provides flexible settings that can be changed via SDR programming software, such as SDRangel, SDR++, sdrtrunk, and many more free options, and even a few paid options for particular use, such as Studio1 and Sodira to name a few. Using SDR software combined with the SDR device best suited for the use case needs, frequency, bandwidth, modulation or demodulation method, gain, and many more can be fine-tuned without needing a mountain of extra hardware, saving on invested costs and time. [2], [7 - 9]

5.1 RTL-SDR

The RTL-SDR is a receiver-only SDR, making its application slightly narrower compared to other devices that are TX and RX capable via duplexing. Applications for this device, particularly those for HAB, are still vast when considering this drawback. Table 3 from the RTL-SDR website lists all frequency ranges this dongle can listen to, including amateur radio and APRS (Automatic Packet Reporting System) frequencies for even longer-range communications, monitoring meteor scatter, and tracking, which can be applied to research data collection. The RTL-SDR Blog V3/V4 also contains built-in upconverters to expand up to the 1.766 GHz band or frequencies below 24 MHz with the most recent improvements to the hardware. The device is capable of 3.2 Msps. However, this tends to be unstable, with 2.56 Msps being a safer, lossless rate, though some reports say that a rate of 2.8 Msps is possible over USB 3.0 connections. [9]



Figure 3: The most recent version of the RTL-SDR Blog RTL-SDR V4 with the Dipole Antenna Kit. Retailing from the official store, this kit is the most cost-effective SDR out of these common devices at under \$50. <https://www.rtl-sdr.com/buy-rtl-sdr-dvb-t-dongles/>

The device originated from Digital Video Broadcasting – Terrestrial (DVB-T) television tuners designed around the RTL2832U chipset, whose raw I/Q data could be directly accessed, which was discovered by Antti Palosaari, Eric Fry, and Steve Markgraf with Osmocom. Markgraf would develop custom software to convert these tuners into a wideband SDR device, and due to the wide availability and ease of use of the tuners, the device has been able to maintain this low price point.[9]

Table 3: Similar tuner devices based off the same hardware used in the RTL-SDR Blog V3/V4 show similar or even particular capabilities. <https://www.rtl-sdr.com/about-rtl-sdr/>

Tuner	Frequency range
Elonics E4000	52 – 2200 MHz with a gap from 1100 MHz to 1250 MHz (varies)
Rafael Micro R820T/2/R860	24 – 1766 MHz (Can be improved to ~13 - 1864 MHz with experimental drivers)
Rafael Micro R828D	RTL-SDR Blog V4 Model: 24 - 1766 MHz. Other models may have a cutoff around 1 GHz.
Fitipower FC0013	22 – 1100 MHz
Fitipower FC0012	22 – 948.6 MHz
FCI FC2580	146 – 308 MHz and 438 – 924 MHz (gap in between)

Table Source: [Osmocom](https://www.osmocom.com/)

5.2 ADALM-PLUTO

5.2.1 Description

The Analog Devices Inc. (ADI) ADALM-PLUTO, or PlutoSDR, is a fully self-contained SDR device with transmit and receive capabilities. This compares to the RTL-SDR, which is receive-only. This device comes at a price point of \$195 USD as noted in *Figure 4*, much higher than the RTL-SDR Blog V4 dongle with the Dipole telescopic antenna kit at just \$50 as in *Figure 3*. The capable range off-the-shelf is from 325 MHz to 3.8 GHz, though with modifications this range can be extended from 70 MHz to 6 GHz, though reports have shown this to be unstable at times. Since this device is self-contained, external power from the device must be supplied via Micro-USB, which also provides a data channel to SDR software of the user's choice, pending proper drivers exist for that software. To compare to the RTL-SDR, five categories will be observed; LoRa band capability, sample rate and resolution, ease of use, software support, and performance from the perspective of H.A.B. applications. [10], [12]

ADALM PLUTO (PlutoSDR)*

Cost: \$99 USD (Special), \$149 USD
Frequency Range: 325 – 3800 MHz (default), 70 - 6000 MHz (with firmware hack)
ADC Resolution: 12 Bits
Max Bandwidth: 20 MHz (default), 56 MHz (with firmware hack)
TX/RX: TX and RX (Full Duplex)
Preselectors: None
Release Date: Mid 2017

The PlutoSDR is a low cost full duplex TX and RX receiver designed by the big silicon company Analog Devices. It is designed mostly for University students to use for learning about RF and SDR concepts, but it can also find use as a general purpose experimenters SDR.



Figure 4: Quick specifications of the ADALM-PLUTO, A.K.A. the PlutoSDR, a TX/RX device made by Analog devices. However, prices have increased since this description was written to \$195 USD. <https://www.rtl-sdr.com/roundup-software-defined-radios/>

5.2.2 LoRa Band Compatibility & Sample Rate

As described in Section 5.2.1 and shown in Figure 4, the frequency range of the PlutoSDR can range from either 325 MHz – 3.8 GHz or 70 MHz – 6 GHz with firmware hacking. This is much different from the 24 MHz – 1.766 GHz, though both easily cover the LoRa frequency bands at 433, 868, and 915 MHz, though the PlutoSDR requires technical updates to operate on lower, commonly used frequencies such as the 2-meter (144-148 MHz) amateur radio band, which while at non-LoRa frequencies, does hamper HAB applications in terms of communication. The capabilities of the PlutoSDR for transmission and receiving allow for similar configurations on the balloon's payload and on the ground, allowing for two-way communication. [10], [11]

Resolution and sampling rate are where these devices begin to differ even more. The maximum sample rate of the PlutoSDR is 61.44 Msps, though without a USB 3.0 connection, this rate is reduced to around 6 Msps, only 1/10th the maximum capable rate. The PlutoSDR has a 12-bit

ADC (Analog to Digital Converter) and DAC (Digital to Analog Converter) built into the device and can easily oversample LoRa channels to provide higher resolution to get better SNR values for weak LoRa signals. This compares to the 8-bit ADC, though the effective number of bits (ENOB) is closer to 7 bits and a maximum sample rate of about 3.2 Msps, with 2.56 Msps of complex samples being a more reliable rate. While this is adequate for LoRa at a typical bandwidth of ~8-500 kHz, the intermediate bandwidth of 3.2 MHz still shows losses for weak-signal dynamic ranges at the 8-bit resolution. [10], [13]

5.2.3 Ease of Use & Support

This is where the PlutoSDR starts to stumble when compared to the RTL-SDR and the HackRF in the next section. As the target client for the PlutoSDR is geared more towards development and learning as a platform, it requires special proprietary libiio drivers from Analog Devices for all operating systems. The device can be accessed via MATLAB, Simulink, or GNU Radio using learning as a platform; it requires special proprietary libiio drivers from Analog Devices for all ADI's proprietary gr-iio blocks with official software support. Some example code is provided; the user will require familiarity with configuring embedded devices, particularly embedded Linux. This compares to the RTL-SDR's plug-and-play ability right out of the box, at least for Windows and Linux platforms. [10]

Both official and community support for the RTL-SDR outweighs the PlutoSDR and is matured with software support for various SDR ecosystems. Among the shared support for MATLAB and Simulink, as well as GNU Radio via `gr-osmosdr` or `SoapyRTLSDR` blocks, require little to no prep for inexperienced users, especially when utilizing Soapy-compatible software, enabling inexperienced users even further. This is contrasted by a lack of community software support and documentation on Analog Devices' website, making this platform difficult despite its open-source firmware. This full-source openness does allow for custom code to directly control the SDR, which is of interest to intermediate users; combined with MATLAB and GNU Radio's community support for the device keeps it a capable device, though for the price difference, it requires far too much in the way of research, previous experience, and firmware tinkering to be viable for, i.e., students with little SDR or radio experience, to use. [10], [14]

5.2.4 Performance & H.A.B Considerations

Signal sensitivity in both devices is high enough for HAB applications using LoRa modulation, but PlutoSDR gains have an edge here. Pluto's front-end AD9363 provides about a 2.5 dB full-gain noise figure, which can yield higher sensitivity for weak LoRa signals. The higher noise floor of the RTL-SDR at roughly 4-5 dB with the 8-bit ADC makes this process more difficult, requiring a detectable signal to be higher. This means the PlutoSDR can detect LoRa packets at slightly farther distances or higher SF values than the RTL-SDR. [15]

The RTL-SDR, particularly the V4, can better handle Doppler effects due to shifts in flight during HAB applications, showing much more stable tuning at one ppm temperature compensated crystal oscillator (TCXO) with roughly ± 1 kHz error at the 915 MHz band, which allows for better carrier tracking. This is not the case in the PlutoSDR, which lacks this TCXO component, reducing accuracy to about 25 ppm and allowing drift in the 10s of kHz, proving a much less stable connection overall. [16]

The RTL-SDR interference tolerance has been noted previously (see section 5.1), including a

triplexer, notch filters that isolate bands, and metal shielding to prevent overload on the receiver. The PlutoSDR lacks these, but the 12-bit range allows the tolerance of stronger out-of-band signals more effectively than the RTL-SDR. Despite that, the RTL-SDR yields much clearer LoRa reception due to better filtering methods. Due to the nature of balloon launches, the payload will often retain line-of-sight for several kilometers, giving both SDR devices difficulty without a proper antenna—the data rates of each device and the throughput limit of the USB limit this. While receiver sensitivity may give the PlutoSDR the edge, either device can perform this function reliably. [13]

5.3 RTL-SDR Vs. HackRF One

5.3.1 Description

Developed by Great Scott Gadgets, this competent device with a range from 1 MHz to 6 GHz, a range that envelopes the other devices and has room for more. While half-duplexed, compared to the PlutoSDR full-duplex setup for the transceiver, the wide range of applications and support close this gap quite a bit. Though this is not without cost, however, as the HackRF One is the most expensive device to be observed, at a price point of nearly \$340, this device falls short of some of the capabilities of the PlutoSDR, but what it can do is covered by a large amount of community support and clear documentation that will make this device friendly to both beginners and intermediates.[17]

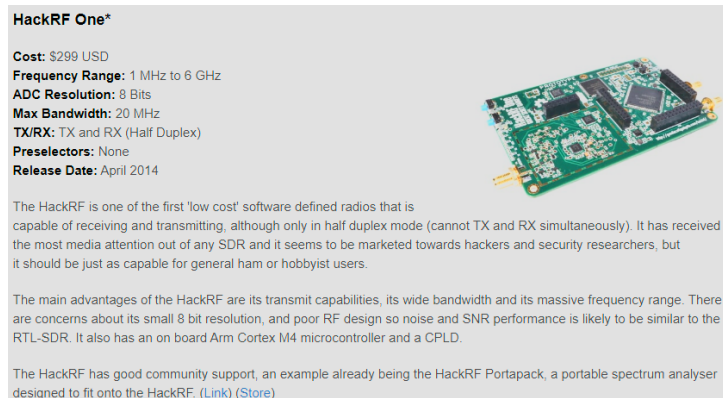


Figure 5: A description of the Great Scott Gadgets' HackRF One, a massively capable SDR module with light comparison to the RTL-SDR module. What it misses in price it makes up for in frequency range and community support. <https://www.rtl-sdr.com/roundup-software-defined-radios/>

5.3.2 LoRa Band Compatibility & Sample Rate

As the HackRF One module is capable of such a broad range of frequencies, it is safe to say that this device is well within compatible for LoRa/LoRaWAN compatibility. The sample rate of the HackRF is low when compared to the PlutoSDR, capable of only 20 Msps as compared to the PlutoSDR's maximum of 61.44 Msps, though still much higher than the RTL-SDR's 3.2 Msps when one considers the 8-bit system on both devices. The HackRF One, because of this high bitrate, is capable of a 20 MHz bandwidth, allowing it to scan multiple channels or even oversample if needed. For a narrowband LoRa transmission, either device is more than capable, though the HackRF shines at broad bandwidths and maintains better resolution than the RTL-SDR. However, it requires more processing on the host machine than an RTL-SDR, which is suboptimal for HAB applications where low power demands are necessary to ensure the longevity of device function for long flights.[9],[17]

5.3.3 Ease of Use & Support

The HackRF One is a significantly more complex device, and at the higher price point, it very well should be. Understanding the device directly out of the box is difficult, to say the least, as one must understand what proper drivers are needed and what toolboxes for SDR programming software are needed, such as the HackRF tools for GNU Radio, such as `hackrf-tools` or the `SoapySDR`. However, both the HackRF One and the RTL-SDR have tremendous documentation and community support with tutorials galore for understanding initial setup and downloading peripherals as needed to get started, allowing amateur hobbyists, intermediates, and researchers alike to access the tools necessary to succeed when using either device. In contrast, hardware additions such as PAs or LNAs are not built into the HackRF module and require external connection via additional wiring, unlike the plug-and-play RTL-SDR, which has a built-in bias-tee, allowing for heatsink and amplifier additions. This beats the HackRF in terms of ease of use, but the HackRF wins in terms of flexibility.[9], [17-18]

Software support for SDR compatibility is equal for these two devices, allowing for a host machine to support the HackRF and the RTL-SDR so long as the drivers are available for the device on the host machine. The proper toolboxes are added to the SDR software for instances like MATLAB, which would require the Communications Toolbox. However, this software has a usage cost, making it less viable for hobbyists. [19]

5.3.4 Performance & H.A.B Considerations

In real high-altitude balloon experiments, either device can receive LoRa signals up to stratospheric altitudes, providing a reasonable antenna/link budget. For example, radiosonde and LoRa balloon projects often use RTL-SDR receivers to capture telemetry and GPS data. One HAB enthusiast reported using an RTL-SDR-based LoRa gateway to recover a payload at 143,000 ft. or roughly 43 km. Similarly, a blog post shows a HackRF decoding pico-balloon transmissions, noting that the "cheaper RTL-SDR could also be used" for tracking. These accounts suggest that both SDRs are feasible for LoRa/HAB work. [20 - 22]

However, practical performance can differ. HackRF's front-end and 8-bit ADC are known to be noisier: one comparative test found HackRF produced the lowest SNR and overloaded easily under strong signals. By contrast, the RTL-SDR V4 was designed with improved filters and a stable TCXO and its tuner (R828D) have good inherent sensitivity. The V4 datasheet cites "improved sensitivity" and notch filters to suppress interference. In HAB tracking, Doppler shift and clock stability matter: the RTL-SDR's one ppm TCXO provides much tighter frequency accuracy than HackRF's ~20 ppm crystal. This means the RTL-SDR will drift far less as the balloon moves. [9], [22-24]

In practice, antenna and decoder quality often dominate. LoRa's chirp modulation can tolerate some frequency error, but major Doppler (tens of kHz at 915 MHz for swift ascent) may require software correction. Both SDRs can feed GNU Radio LoRa demod blocks (e.g., `gr-lora`) or other LoRa decode libraries. The built-in bias-tee on the RTL-SDR V4 makes powering an LNA easy, which boosts weak signals at great range. The HackRF can also employ external amplifiers, but without a bias-tee, one must supply them separately.

In summary, the LoRa HAB range is limited more by antenna gain and payload power than the

SDR choice. Both SDRs have successfully captured balloon LoRa links (~tens of km), but RTLSDR V4 generally offers cleaner reception (higher SNR and stable tuning). HAB chasers more commonly use it.

6. Controlled Transmission Experiments on Changing Receivers

In a controlled experiment testing the hardware reception performance of three standard software-defined radios (SDRs) using a 915 MHz LoRa transmission, notable differences emerged among the HackRF One, PlutoSDR, and RTL-SDR V4. The transmission was carried out using the HackRF One with a spread factor of 9, a base bandwidth of 125 kHz, and an ANT500 telescopic antenna. The signal gain was standardized with a 30 dB power amplifier on the transmitter and a 30 dB LNA on the weakest receiver to ensure comparability. Operating at a sample rate of 2.5 Msps, the HackRF One also performed as a moderate receiver, introducing minor noise through signal amplification but still delivering consistent symbol reception. The average RSSI was measured at -30 dB, with most LoRa chirps visible and demodulated, validating its viability as a mid-tier option in both transmission and reception roles. In *Figure 6*, the top left plot is the transmission spectrum. Below, on the bottom left of the image, is the waterfall plot, where the bright center line is the carrier and the short bursts are the chirp signal. The settings are on the top right of the image, which also displays a status and decoded message received, and finally, on the bottom right is the channel power received, which also displays RSSI.

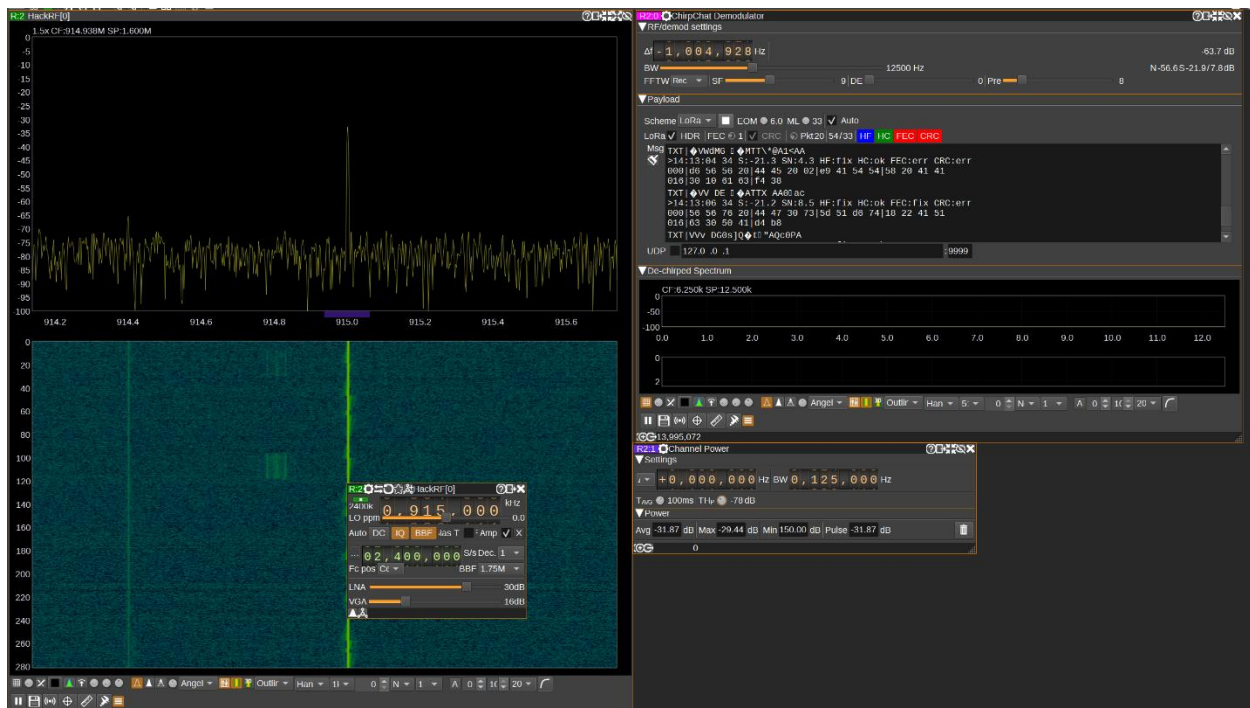


Figure 6: SDRangel data related to reception on the HackRF One device. Note the LoRa demodulator "Msg" window data has "?" symbols.

In contrast, the PlutoSDR demonstrated significant limitations in the same setup. Despite operating at the same distance, gain, and sample rate, the PlutoSDR registered a much lower average RSSI of -54 dB. While the chirp signal was faintly visible on the waterfall plot in *Figure 7*, no symbols were successfully received or decoded during the test. This underperformance suggests that the PlutoSDR's reception is sensitive to reduced power scenarios and may require optimized external power supply or driver-level adjustments to function reliably under constrained conditions. Its internal architecture and software dependencies likely hinder its performance without ideal conditions. This is critical for high-altitude ballooning (HAB) or low-power field applications where energy availability is limited.

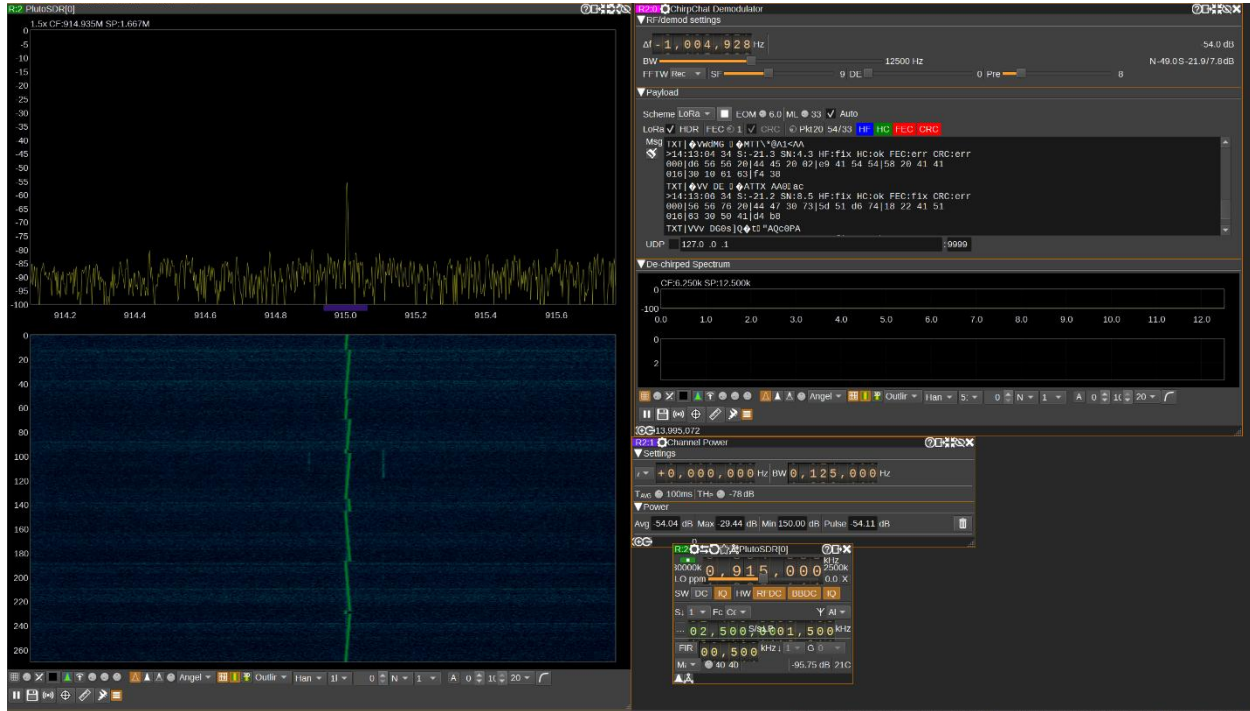


Figure 7: Similar to Figure 6, this image displays the reception on the PlutoSDR.

The RTL-SDR V4, though the most budget-friendly device tested, delivered a better performance than the PlutoSDR in this scenario. While the received signal was noisier and the chirp bandwidth appeared broader, the device achieved an RSSI of -47 dB, about 7 dB stronger than the PlutoSDR as noticed in *Figure 8*. Though it lacked stable symbol reception, the RTL-SDR's reduced power consumption (approximately 1W via USB 2.0) and minimal hardware requirements position it as a practical choice for cost-sensitive, power-constrained missions. While internal amplification is limited, the device could be paired with an external LNA and remain well below the total cost of the PlutoSDR or HackRF One. These results highlight the trade-offs between signal fidelity, power requirements, and budget, particularly critical factors in remote or airborne LoRa deployments.

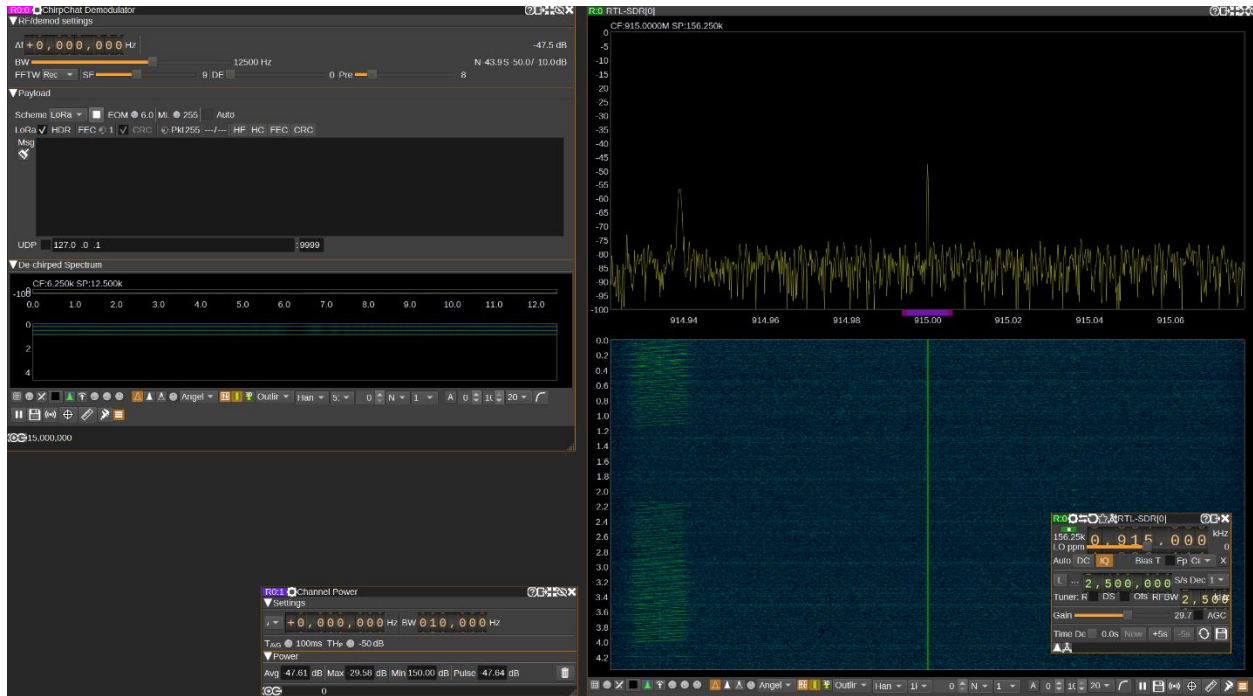


Figure 8: Reception plots from the RTL-SDR V4 dongle. An important observation is the chirp signal received to the left of the center band, where the signal is much more visible, but noise distorts the waterfall plot much more than previously.

7. Conclusions

For HAB research specifically, recent work has validated LoRa at extreme altitudes in an experiment by the University of Grenoble and partners. LoRa signals were tested on balloons up to ~30 km altitude and >500 km range. They found that LoRa's links remain robust in near-space conditions, and satellite integration can extend coverage. LoRa's robustness to Doppler (due to chirp modulation) and ability to tolerate link shadowing make it well-suited for high-velocity, high-altitude. In general, LoRa/LoRaWAN has become a technology of choice for any application demanding long-range and low power, from remote weather stations to Internet-of-Things prototypes.

LoRa (the PHY) provides a flexible chirp-spread spread-spectrum modulation where bandwidth, spreading factor, and coding rate can be tuned for the required range and throughput. LoRaWAN (the network) manages devices and frequencies in an IoT network. The interplay of SF and BW governs the trade-off between robustness (range, link budget) and data rate (throughput, battery life). In research and prototyping, various SDR hardware (like RTL-SDR, PlutoSDR, and HackRF) can generate and receive LoRa signals, each with its own spectrum coverage and cost. These SDRs enable rapid experimentation with different LoRa parameters and antennas, which is valuable for HAB payload development and broad IoT system design and testing. Can software-defined radio (SDR) platforms be used to analyze, develop, and deploy LoRa and LoRaWAN systems? Yes, and with additional enhancer equipment such as external power amplifiers and low-noise amplifiers, and the proper antenna, LoRa and LoRaWAN are a more than viable option for the development of SDR-based communication systems for HAB experiments.

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